

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **SPCL_GTM-03 (FN)** **Date: 13.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Three equal weights A, B and C of mass 2 kg each are hanging on a string passing over a fixed frictional pulley as shown in the figure. The tension in the string connecting weight B and C is approximately $\left(\text{Take } g = 10\, m/s^2\right)$

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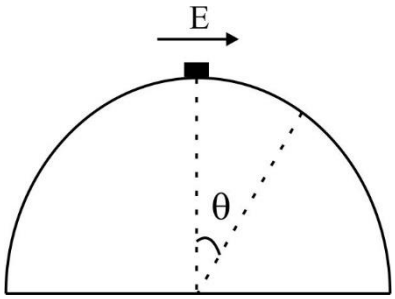


A) zero B) 13.3 N C) 3.3 N D) 19.6 N

2. Eight spherical rain drops of the same mass and radius are falling down with a terminal speed of $6 cms^{-1}$. If they coalesce to form one big drop, its terminal speed will be (Neglect the buoyancy due to air)

A) $1.5 cms^{-1}$ B) $6 cms^{-1}$ C) $24\ cms^{-1}$ D) $32\ cms^{-1}$

3. A positively charged small disc is released from the top of fixed hemispherical frictionless dome in presence of a uniform horizontal electric field of magnitude E as shown in figure. If the disc leaves the dome after an angular displacement $\theta = \sin^{-1}(3/5)$, find the ratio of gravitational and electrostatic forces on the disc. Assume that the dome does not exhibit any electrical property.



A) $\frac{4}{3}$ B) $\frac{3}{4}$ C) $\frac{9}{2}$ D) 12

4. The temperature at which the velocity of sound in oxygen will be the same as that of nitrogen at $15^0 C$ is

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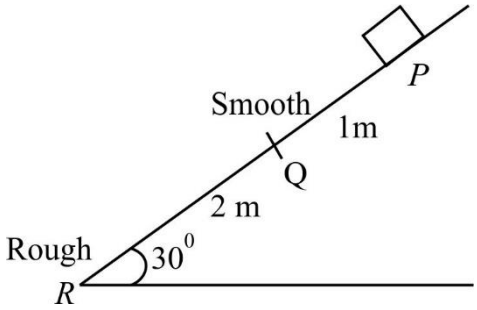
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A) 112^0C B) 72^0C C) 56^0C D) 17^0C

5. A triangular loop of side l carries a current I. It is placed in a magnetic field B such that the plane of the loop is in the direction of B. The torque experienced by the loop is

A) zero B) IBl C) $\frac{\sqrt{3}}{2}Il^2B^2$ D) $\frac{\sqrt{3}}{4}IBl^2$

6. A block of mass 5.0 kg slides down from the top of an inclined plane of length 3 m. The first 1m of the plane is smooth and the next 2m is rough. The block is released from the rest and again comes to rest at the bottom of the plane. If the plane is inclined at 30^0 with the horizontal, the coefficient of friction for the rough portion.



A) $\frac{2}{\sqrt{3}}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{\sqrt{3}}{4}$ D) $\frac{\sqrt{3}}{5}$

7. An electrical cable of copper has just one wire of radius 9mm. Its resistance is 5 Ohm. This single copper wire of the cable is replaced by 6 different well insulated copper wires of same length in parallel each of radius 3mm. The total resistance of the cable will now be equal to

A) 7.5 Ohm B) 45 Ohm C) 90 Ohm D) 270 Ohm

8. Two open organ pipes of fundamental frequencies n_1 and n_2 are joined in series. The fundamental frequency of the new pipe so obtained will be

A) n_1+n_2 B) $\frac{n_1n_2}{n_1+n_2}$ C) $\frac{n_1+n_2}{2}$ D) $\sqrt{\left(n_1^2+n_2^2\right)}$

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9. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of the molecular weight of A and B is

A) 2 B) 1/2 C) 2/3 D) 3/4

10. The acceleration due to gravity on a planet is same as that on Earth and its radius is four times that of Earth. The value of escape velocity on that planet if it is v_e on the surface of Earth is___

A) v_e B) $2v_e$ C) $4v_e$ D) $\frac{v_e}{2}$

11. A plane electromagnetic wave of intensity $10\,W/m^2$ strikes a small mirror of area $20cm^2$, held perpendicular to the approaching wave. The radiation force on the mirror will be

A) $6.6\times10^{-11}N$ B) $1.33\times10^{-11}N$ C) $1.33\times10^{-10}N$ D) $6.6\times10^{-10}N$

12. When the angle of incidence on a material is 60^0, the reflected light is completely polarized. The velocity of the refracted ray inside the material is $\left(\text{in } ms^{-1}\right)$

A) 3×10^8 B) $\left(\frac{3}{\sqrt{2}}\right)\times10^8$ C) $\sqrt{3}\times10^8$ D) 0.5×10^8

13. In a metre bridge experiment null point is obtained at 20cm from one end of the wire when resistance X is balanced against another resistance Y. If $X<Y$, then where will be the new position of the null point from the same end, if one decided to balance a resistance of $4X$ against Y

A) 50 cm B) 80 cm C) 40 cm D) 70 cm

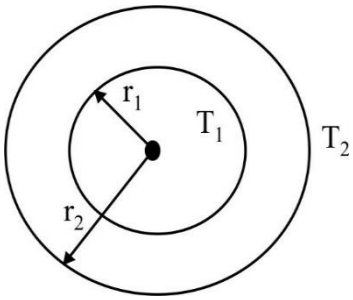
14. The length of a simple pendulum is about 100cm known to have an accuracy of 1mm. Its period of oscillation is 2s determined by measuring the time for 100 oscillations using a clock of 0.1s resolution. The accuracy in the determined value of g is____

A) 0.2% B) 0.5% C) 0.1% D) 2%

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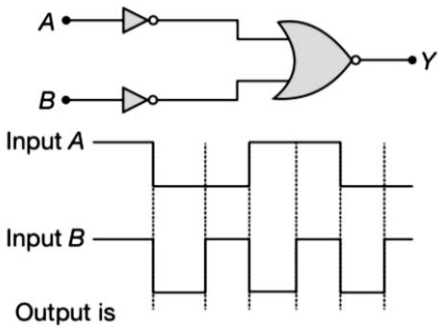
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15. The figure shows a system of two concentric spheres of radii r_1 *and* r_2 kept at temperature T_1 *and* T_2 respectively. The radial rate of flow of heat in the substance between the concentric spheres is proportional to

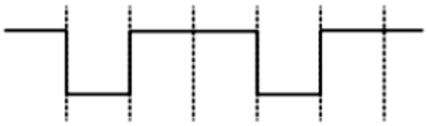


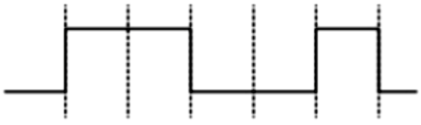
A) $\frac{r_1 r_2}{r_2 - r_1}$ B) $(r_1 - r_2)^2$ C) $\frac{r_1 - r_2}{r_1 r_2}$ D) $\log\left(\frac{r_2}{r_1}\right)$

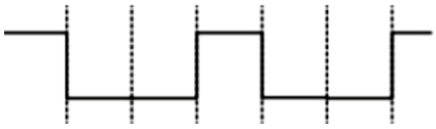
16. The logic circuit shown below has the input wave form A and B as shown. Pick out the correct output waveform



A) 

B) 

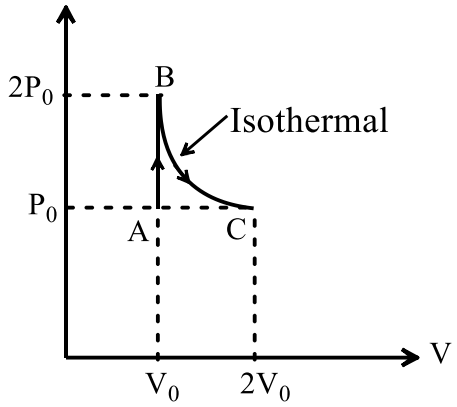
C) 

D) 

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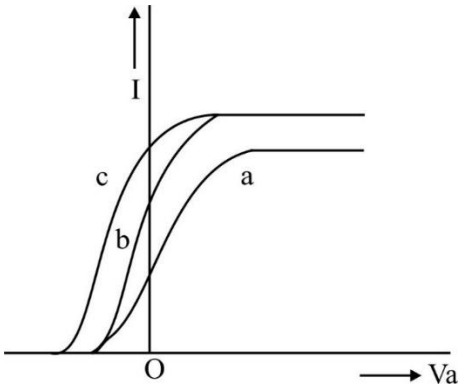
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17. A diatomic ideal gas undergoes a thermodynamic change (from A to B and then from B to C) according to the $P-V$ diagram shown in the figure. The total heat given to the gas is nearly (use $\ln 2 = 0.7$)



A) $2.5P_0V_0$ B) $1.4P_0V_0$ C) $3.9P_0V_0$ D) $1.1P_0V_0$

18. The figure shows the variation of photocurrent (I) with anode potential (V_a) of a photosensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b *and* f_c the frequencies for the waves a, b and c respectively



A) $f_a = f_b$ *and* $I_a \neq I_b$ B) $f_a = f_c$ *and* $I_a = I_c$

C) $f_a = f_b$ *and* $I_a = I_c$ D) $f_b = f_c$ *and* $I_b = I_c$

19. A wind powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of wind energy intercepted by its blades into

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electrical energy. For wind speed v, the electrical power output will be proportional to

A) v B) v^2 C) v^3 D) v^4

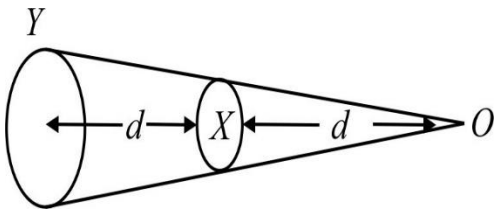
20. If $\vec{a}$ and $\vec{b}$ are two unit vectors and $\vec{R}=\vec{a}+\vec{b}$ and also if $|\vec{R}|=R$, then

A) R < 0 B) R > 2 C) $0 \le R \le 2$ D) R must be 2

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Two circular coils X and Y, having equal number of turns and carrying currents in the same sense subtend same solid angle at point *O*. If the smaller coil *X* is midway between *O* and *Y* and if we represent the magnetic induction due to bigger coil *Y* at *O* as B_y and that due to smaller coil *X* at *O* as B_x, then find the ratio B_x / B_y.

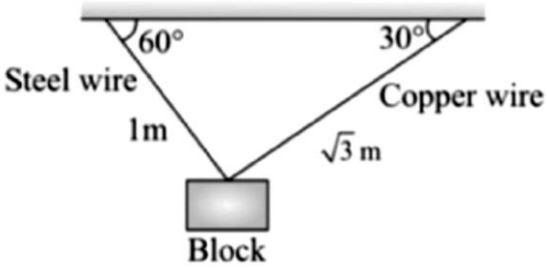


22. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area $0.5cm^2$ and length $\sqrt{3}m$ and $1m$, respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30^0 and 60^0 respectively. If elongation in copper wire is (Δl_C) and elongation in steel wire is (Δl_S), then the ratio $\frac{\Delta l_C}{\Delta l_S}$ is _____

[Young's modulus for copper and steel are $1\times10^{11}N/m^2$ and $2\times10^{11}N/m^2$ respectively]

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23. The flux linked with a coil of 100 turns varies as $\phi=\begin{cases}\phi_m\left(1-\frac{4t}{T}\right); & 0\le t\le\frac{T}{2}\\ \phi_m\left(\frac{4t}{T}-3\right); & \frac{T}{2}\le t\le T\end{cases}$

Where $\phi_m=0.02Wb$ and $T=\frac{1}{50}s$. Then the maximum magnitude of induced emf (in V) is____

24. When ${}_{92}U^{235}$ undergoes fission, about 0.1% of the original mass is converted into energy. Then the amount of ${}_{92}U^{235}$ should undergo fission per day in a nuclear reactor so that it provides 200 megawatt electric power is $96P\times10^{-3}kg$ then $P=$____

25. A particle moves according to the law $x=a\cos\frac{\pi t}{2}$. The distance covered by it in the time interval between $t=0$ to $t=3s$ is xa. Find the value of 'x'.

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CHEMISTRY **MAX.MARKS: 100**

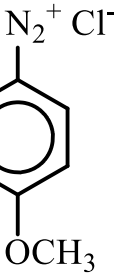
SECTION – I

(SINGLE CORRECT ANSWER TYPE)

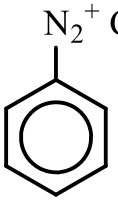
This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The correct reactivity order of the following diazonium salt towards coupling reaction with phenol (in basic medium) is

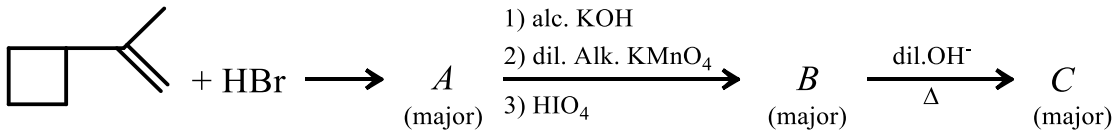
a) 

b) 

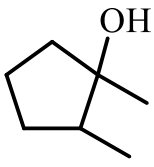
c) 

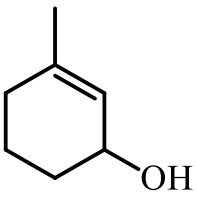
d) 

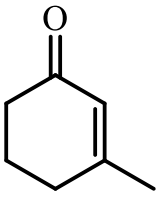
A) $a > b > c > d$ B) $b > c > a > d$ C) $b > d > c > a$ D) $c > a > b > d$

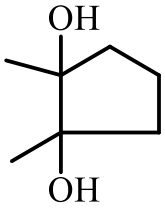
27. 

Select the compound C among the following

A) 

B) 

C) 

D) 

28. Which of the following represents the correct increasing order of first ionisation enthalpy for *Ca, Ba, S, Se, Ar*?

A) $Ba < Ca < Se < S < Ar$ B) $Ca < Ba < S < Se < Ar$

C) $Ca < S < Ba < Se < Ar$ D) $S < Se < Ca < Ba < Ar$

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29. Select the correct set of reactions and reagents from the following

List-I (Reaction)		List-II (Reagent)	
A)	Hoffmann degradation	(I)	Con. KOH, Δ
B)	Clemmensen reduction	(II)	$CHCl_3, NaOH$ / H_3O^+
C)	Cannizaro reaction	(III)	$Br_2, NaOH$
D)	Reimer-Tiemann reaction	(IV)	$Zn-Hg$ / HCl

A) $A-III; B-IV, C-II, D-I$ B) $A-II; B-IV, C-I, D-III$

C) $A-III; B-IV, C-I, D-II$ D) $A-II; B-I, C-III, D-IV$

30. Select the option with correct property

A) $\left[Fe(CN)_6\right]^{4-}$ and $\left[Fe(CN)_6\right]^{3-}$ both are paramagnetic

B) $\left[Fe(CN)_6\right]^{4-}$ and $\left[Fe(CN)_6\right]^{3-}$ both are diamagnetic

C) $\left[Fe(CN)_6\right]^{3-}$ is paramagnetic, $\left[Fe(CN)_6\right]^{4-}$ is diamagnetic

D) $\left[Fe(CN)_6\right]^{3-}$ is diamagnetic, $\left[Fe(CN)_6\right]^{4-}$ is paramagnetic

31. **Statement-I:** Solid $AlCl_3$ is a covalent compound and form ionic solution in water

Statement-II: BCl_3 does not undergo hydrolysis

A) Both statements I and statement II are true

B) Statement I is true but statement II is false

C) Statement I is false but statement II is true

D) Both statements I and statement II are false.

32. Choose the correct statements from the following:

a) H_2S act as reducing agent while SO_2 can act as a reducing and oxidising agent

b) SO_2 acts as bleaching agent but not SO_3

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c) order of acidic nature $SO_3 < SO_2$

d) Number of lone pair of electrons in O_2 molecule is 8 according to VBT

A) a and b only B) b and c only C) b and d only D) c and d only

33. 19 gm of a mixture containing $NaHCO_3$ and Na_2CO_3 on complete heating liberates 1.12 lit of $CO_{2(g)}$ at NTP. What is the weight (in grams) of Na_2CO_3 in the mixture before heating?

A) 10.6 B) 5.3 C) 4.2 D) 8.4

34. **Statement-I:** $Na_2S_2O_3$ undergoes oxidation to form $Na_2S_4O_6$ in the presence of I_2

Statement-II: Cl_2 undergoes disproportionation to form Cl^- and ClO_3^- in the presence of hot concentrated solution of $NaOH$

A) Statement I is true but statement II is false

B) Statement I is false but statement II is true

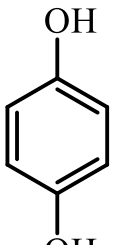
C) Both statement I and statement II are true

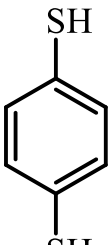
D) Both statement I and statement II are false

35. Number of molecules with zero net dipole moment from the following $(\mu_{net} = 0)$

(1) CH_2Cl_2 (2) CCl_2Br_2 (3) PF_3Cl_2 (4) SF_4

(5) XeF_6 (6) XeO_3 (7) XeO_2F_2 (8) CO

(9) 

(10) 

A) 4 B) 3 C) 0 D) 1

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36. Match the following complexes in List-I with the configuration in List-II of the metal ion in the complex.

List-I Complex compound		List-II Electronic configuration	
a)	$\left[Co\left(C_2O_4\right)_3\right]^{3-}$	(*p*)	$e_g^4\, t_{2g}^4$
b)	$\left[Fe\left(CN\right)_6\right]^{3-}$	(*q*)	$t_{2g}^6\, e_g^0$
c)	$\left[CoCl_6\right]^{3-}$	(*r*)	$t_{2g}^5\, e_g^0$
d)	$\left[NiCl_4\right]^{2-}$	(*s*)	$t_{2g}^4\, e_g^2$

A) $a-q;b-s;c-p,d-r$ B) $a-q;b-r;c-s,d-p$

C) $a-r;b-q;c-s,d-p$ D) $a-s;b-q;c-r,d-p$

37. Correct set of Quantum number for the outermost electron in Cr

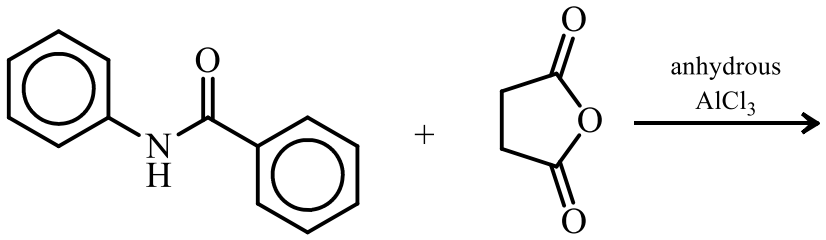
A) $n=3$ $l=2$ $m=-2$ $s=+\frac{1}{2}$

B) $n=4$ $l=0$ $m=1$ $s=+\frac{1}{2}$

C) $n=3$ $l=1$ $m=0$ $s=+\frac{1}{2}$

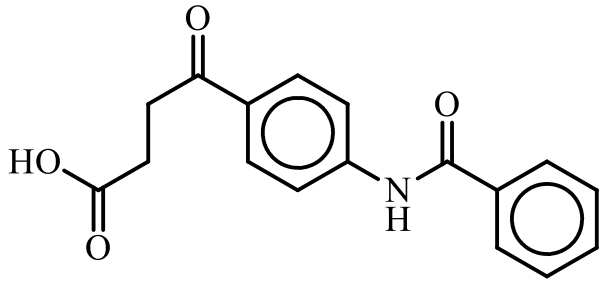
D) $n=4$ $l=0$ $m=0$ $s=+\frac{1}{2}$

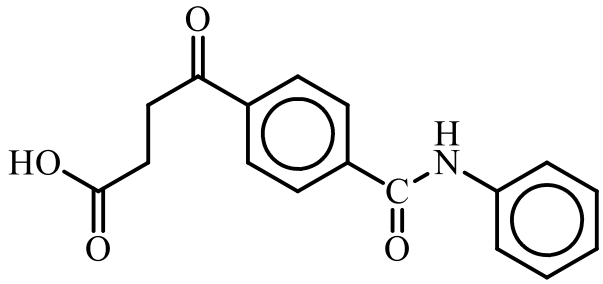
38. What is the major product obtained in the following reaction?

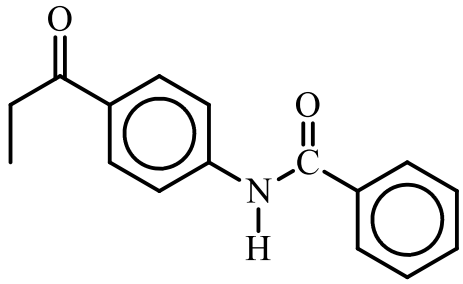


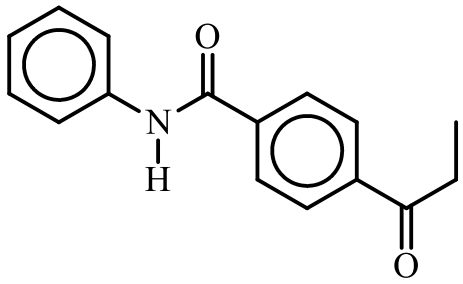
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A) 

B) 

C) 

D) 

39. Identify the correct statements

a) Mn_2O_7 is an acidic oxide

b) Cr_2O_3 is an amphoteric oxide

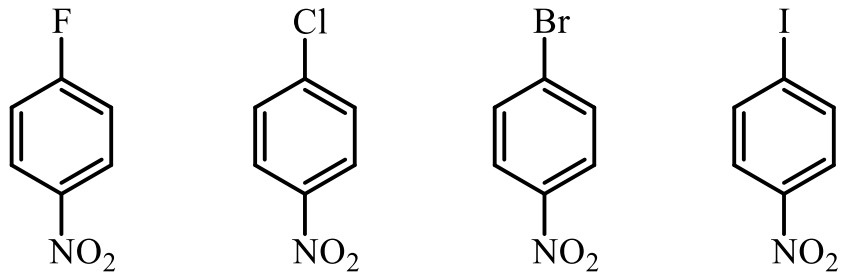
c) NO and N_2O are neutral oxides

d) CO_2 and SiO_2 are acidic oxides

A) a and b only B) a and c only C) a, b, c and d D) b, c and d only

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40. What is the correct order of the following compounds towards S_NAr reaction (Nucleophilic addition – elimination)



(I) (II) (III) (IV)

A) $(IV)>(III)>(II)>(I)$ B) $(I)>(II)>(III)>(IV)$

C) $(I)=(II)=(III)=(IV)$ D) $(IV)>(III)>(I)>(II)$

41. The resistivity of 0.4 M solution of an electrolyte is $10\times10^{-3} ohm-cm$ its molar conductivity is $x\times10^4 ohm^{-1}cm^2mol^{-1}$, then the value of '$x$'is ___

A) 20 B) 25 C) 50 D) 75

42. Number of moles of H^+ ions required by 1 mole of MnO_4^- to oxidises iodide into I_2 is

A) 5 B) 16 C) 3 D) 8

43. An organic compound (X) molar mass 110 gm/mol undergoes esterification with acetic anhydride to give product (ester) with molar mass 194 gm/mol. The number of hydroxyl groups in the compound (X) is_____

A) 5 B) 4 C) 2 D) 3

44. $2CO_{2(g)} \rightleftharpoons 2CO_{(g)} + O_{2(g)}$; $K_p = 0.4 atm$

If two moles each of the gases are present at equilibrium, select the correct statement

A) Reaction is more favourable in forward direction

B) At the given stage the total pressure is 1.2 atm

C) Partial pressure of all gases are not equal at equilibrium

D) Addition of CO gas shifts the equilibrium in forward direction

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45. Which of the following equations describe correct work done during a reversible adiabatic expansion of an ideal gas.

a) $-nRT_1\ell n\left(\frac{V_2}{V_1}\right)$ b) $nC_v(T_2-T_1)$ c) $\frac{nR}{(\gamma-1)}(T_2-T_1)$ d) $(P_1-P_2)(V_2-V_1)$

P_1 and P_2 : Initial and final pressures

V_1 and V_2 : Initial and final volumes

T_1 and T_2 : Initial and final temperatures

n: number of moles of gas

A) a and d only B) b and c only C) a and b only D) c and d only

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The number of monochloro products obtained during the reaction of 2,3-dimethyl butane with Cl_2 in presence of sunlight is (do not consider optical isomerism)

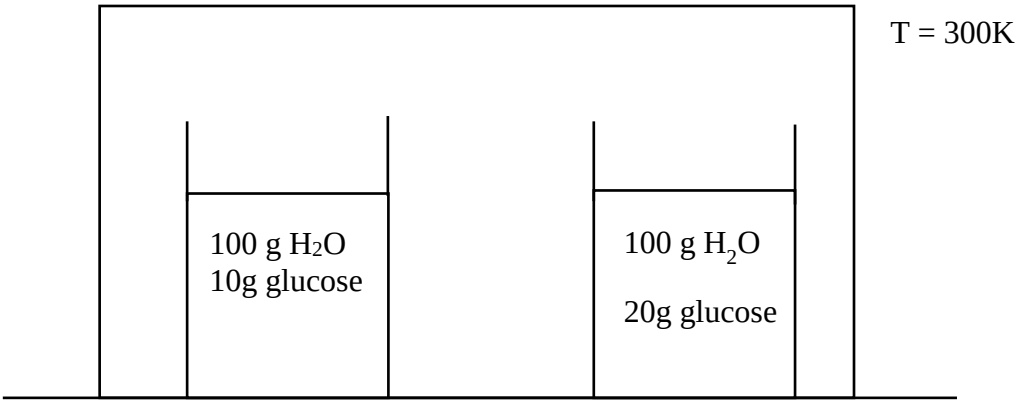
47. Two substance A and B decomposes through 1st order kinetics. If their half lives are 17min and 51 min respectively, then the time taken for concentration of B to be four times that of A, when both start to decompose simultaneously with same initial concentration (in minutes)

48. 5 moles of an ideal gas expands from 10 lit to 20 lit at 300K under isothermal reversible condition. The change in entropy of surroundings in calories is 'X' (consider sign) then the value of $(X+7)$ is_____ (R=2.0 cal/mol/K and $\ln 2 = 0.7$)

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49. When an empty container is inverted on two smaller containers having aqueous solution of glucose as shown below



If the weight of water transferred between the two containers is xg at equilibrium, then the value of $3x$ is (nearest integer) _____

50. Both glucose and fructose undergo cleavage reaction using HIO_4 (Periodic acid). Upon complete oxidation, the number moles of formic acid produced per mole of glucose is X and per mole of fructose is Y. The absolute value of X-Y is_____

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3, \ldots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Which of the following is TRUE?

A) $T_{20} = 1604$ B) $\sum_{k=1}^{20} T_k = 10510$ C) $T_{30} = 3455$ D) $\sum_{k=1}^{30} T_k = 35610$

52. If a, b, c are positive real numbers such that a > b > c and the quadratic equation $(a+b-2c)x^2 + (b+c-2a)x + (c+a-2b) = 0$ has a root in the interval (-1, 0), then the equation $x^2 + (a+c)x + 4b^2 = 0$

A) has both roots equal B) cannot have real roots

C) has real and distinct roots D) none of these

53. Let $f : R \to R$ be a function such that $f(2+x) = f(2-x)$ and $f(7+x) = f(7-x)$ for all real numbers x. If $f(0) = 0$ and there are atleast 'm' number of integer solutions for $f(x) = 0$ in the interval [–2010, 2010], then 'm' can be____

A) 805 B) 705 C) 605 D) 505

54. Let $z = \left(\frac{\sqrt{3}}{2}\right) - i\left(\frac{1}{2}\right)$. The smallest positive integer n such that $\left(z^{95} + i^{67}\right)^{94} = z^n$ is_____

A) 12 B) 10 C) 9 D) 8

55. $\int \frac{\sin\frac{7x}{2}}{\sin\frac{x}{2}} dx$ is equal to (where c is a constant of integration)

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A) $\frac{2\sin 3x}{3}+\sin 2x+2\sin x-x+c$ B) $\frac{2\sin 3x}{3}+\sin 2x-2\sin x+x+c$

C) $\frac{2\sin 3x}{3}-\sin 2x+2\sin x+x+c$ D) $\frac{2\sin 3x}{3}+\sin 2x+2\sin x+x+c$

56. If $P=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}$, $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$, $Q=PAP^T$ and $X=P^TQ^{2010}P$, then X is equal to

A) $\begin{bmatrix}4+2010\sqrt{3} & 8015\\ 2010 & 4-2010\sqrt{5}\end{bmatrix}$ B) $\begin{bmatrix}2010 & 2-\sqrt{3}\\ 2+\sqrt{3} & 2010\end{bmatrix}$

C) $\begin{bmatrix}2+\sqrt{3} & 1\\ 2-\sqrt{3} & 1\end{bmatrix}$ D) $\begin{bmatrix}1 & 2010\\ 0 & 1\end{bmatrix}$

57. If the equations $2x+3y+1=0, 2x+y-1=0$ and $ax+2y-b=0$ are consistent then

A) $a-b=2$ B) $a+b+1=0$ C) $a+b=3$ D) $a-b-8=0$

58. If $A=\begin{bmatrix}a & x & y\\ x & b & z\\ y & z & c\end{bmatrix}$ where $a,b,c,x,y,z\in\{1,2,3,4,5,6\}$ and also a,b,c,x,y,z are distinct, then the number of matrices in A whose sum of diagonal entries is 10 are_____

A) $3(3!)^2$ B) $2(3!)^2$ C) $(3!)^2$ D) $(3!)^3$

59. If $a>0, b>0$ and $a^2+b=2$, then the maximum value of the term independent of x in the expansion of $\left(ax^{\frac{1}{6}}+bx^{-\frac{1}{3}}\right)^9$ is

A) 48 B) 84 C) 42 D) 168

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60. If $f(x+y)=f(x)+f(y)$ for all $x, y \in R$ and $f(1)=1$, then $\lim\limits_{x\to 0}\dfrac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2 f(\sin x)}$ is equal to

A) $\log_e 2$ B) $\log_e 4$ C) $\log_e \sqrt{2}$ D) $\log_e 8$

61. The value of $\sqrt{\pi\left(\int_0^{2008} x|\sin \pi x|\,dx\right)}$ is equal to

A) $\sqrt{2008}$ B) $\pi\sqrt{2008}$ C) 1004 D) 2008

62. A function $y=f(x)$ satisfies the differential equation $f(x).\sin 2x-\cos x+\left(1+\sin^2 x\right)f'(x)=0$ with initial condition $y(0)=0$. The value of $f\left(\dfrac{\pi}{6}\right)$ is equal to

A) $\dfrac{1}{5}$ B) $\dfrac{3}{5}$ C) $\dfrac{4}{5}$ D) $\dfrac{2}{5}$

63. If $A\left(\dfrac{\sin\alpha}{3}-1, \dfrac{\cos\alpha}{2}-1\right)$ and $B(1,1)$, $\alpha \in [-\pi, \pi]$ are two points on the same side of the line $3x-2y+1=0$, then exhaustive set of value(s) of α is_____

A) $\left[-\pi, -\dfrac{3\pi}{4}\right)\cup\left(\dfrac{\pi}{4}, \pi\right]$ B) $[-\pi, \pi]$

C) ϕ D) $\left(0, \dfrac{\pi}{4}\right)$

64. A light ray gets reflected from $x=-2$. If the reflected ray touches the circle $x^2+y^2=4$ and the point of incidence is $(-2,-4)$, then the equation of the incident ray is

A) $4y+3x+22=0$ B) $3y+4x+20=0$

C) $4y+2x+20=0$ D) $y+x+6=0$

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65. $x-2y+4=0$ is a common tangent to $y^2=4x$ and $\frac{x^2}{4}+\frac{y^2}{b^2}=1$. The value of b and the other common tangent are given by

A) $b=\sqrt{3};\ x+2y+4=0$ B) $b=3;\ x+2y+4=0$

C) $b=3;\ x-2y-4=0$ D) $b=\sqrt{3};\ x-2y-4=0$

66. If $\left(\sin^{-1}a\right)^2+\left(\cos^{-1}b\right)^2+\left(sec^{-1}c\right)^2+\left(\cos ec^{-1}d\right)^2=\frac{5\pi^2}{2}$, then the value of $\left(\sin^{-1}a\right)^2-\left(\cos^{-1}b\right)^2+\left(\sec^{-1}c\right)^2-\left(\cos ec^{-1}d\right)^2$ is

A) $-\pi^2$ B) $-\frac{\pi^2}{2}$ C) 0 D) $\frac{\pi^2}{2}$

67. The mean of two samples of sizes 200 and 300 were found to be 25, 10 respectively and their standard deviations are 3 and 4 respectively. The variance of combined sample of size 500 is

A) 64 B) 65.2 C) 67.2 D) 64.2

68. If the line $x=y=z$ intersect the line $(\sin A)x+(\sin B)y+(\sin C)z-2d^2=0=(\sin 2A)x+(\sin 2B)y+(\sin 2C)z-d^2$, then $\sin\left(\frac{A}{2}\right).\sin\left(\frac{B}{2}\right).\sin\left(\frac{C}{2}\right)$ is equal to (where $A+B+C=\pi$)

A) $\frac{1}{8}$ B) $\frac{1}{16}$ C) $\frac{1}{32}$ D) $\frac{1}{12}$

69. Let S and R be two events such that $P\left(\overline{S\cup R}\right)=1/6,\ P\left(S\cap R\right)=1/3$ and $P\left(\overline{S}\right)=1/3$ then S and R events are

A) Independent but not equally likely

B) Independent and equally likely

C) Mutually exclusive and Independent

D) Equally likely but not Independent

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70. If $f(x)$ is a real valued bijective function satisfying $f'(x)=\sin^2(\sin(x+1))$ and $f(0)=3$, then the value of $\left(f^{-1}\right)''(3)$ is equal to:

A) $-\frac{2\sin(\cos1)\sin1}{\sin^5(\cos1)}$

B) $-\frac{2\cos(\sin1)\cos1}{\sin^5(\sin1)}$

C) $-\frac{2\sin(\cos1)\sin^2 1}{\sin^6(\cos1)}$

D) $-\frac{\sin^2(\sin1)}{\cos^2(\cos1)}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. If $\vec{\alpha}$ and $\vec{\beta}$ are two perpendicular unit vectors such that $\vec{x}=\vec{\beta}-(\vec{\alpha}\times\vec{x})$, then the value of $4|\vec{x}|^2$ is

72. Consider the curve $y=\tan^{-1}x$ and a point $A\left(1,\frac{\pi}{4}\right)$ on it. If the variable point $P_i(x_i,y_i)$ moves on the curve for $i=1,2,3.....,n(n\in N)$ such that $y_n=\sum_{m=1}^{n}\tan^{-1}\left(\frac{1}{2m^2}\right)$ and let M_n be slope of line joining A and P_n then $\lim_{n\to\infty}\frac{1}{M_n}$ is.

73. If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{16}+\frac{y^2}{9}=1$ and the hyperbola, $\frac{x^2}{9}-\frac{y^2}{16}=1$ respectively and (e_1,e_2) is a point on the ellipse, $16x^2+9y^2=k$ then k is equal to

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74. Let $f(x)=\begin{cases}\sqrt{x^2-1}, & x\le\sqrt{10}\\ \left(\sqrt{10}x-7\right), & \sqrt{10}<x<5\\ \sin\pi x, & 5\le x<6\\ \{x\}, & 6\le x\le 7\end{cases}$. The number of points where $f(x)$ is discontinuous in $[1,7]$ is: (where {.} denotes fractional part of x.)

75. If the feet of perpendiculars from point $P(4,4)$ to the straight lines $3x+4y+5=0$ and $y=mx+7$ are Q and R respectively, then the value of $3m$, for which area of ΔPQR is maximum, is.

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